PERFECT OBJECTS ARE EXACTLY THE PSEUDO-COHERENT OBJECTS OF LOCALLY FINITE TOR DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the local strictly perfect model for the first implication with pseudo-coherence by locality and flatness of direct summands, and the local tor amplitude bound with the precise criterion for the second.

Editorial note. The label lemma-perfect names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Strictly perfect complexes as local models. Same chapter, the definition of a perfect object through local representatives by strictly perfect complexes. Supplies the starting point of the implication (1) implies (2).
- Pseudo-coherence of strictly perfect representatives and flatness of direct summands. Same chapter, the locality lemma for pseudo-coherence and the tor amplitude lemma for direct summands of finite free modules. Supplies pseudo-coherence and finite tor dimension of the local models in the first implication.
- The precise criterion for perfection. Same chapter, the lemma stating that tor amplitude in a bounded interval plus (a-1)-pseudo-coherence forces perfection. Supplies the second implication once the local tor amplitude bounds are available.

Source reference. Original reference 08CP: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12774–12774. Prerequisite pointer invoked at proof line(s) 12843; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08CC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12066–12066. Prerequisite pointer invoked at proof line(s) 12833; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0654: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12432–12432. Prerequisite pointer invoked at proof line(s) 12836; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let E be an object of $D(\mathcal{O})$. The following are equivalent*

(1) *E is perfect, and*
(2) *E is pseudo-coherent and locally has finite tor dimension.*

Proof. Assume (1). Let U be an object of $\mathcal{C}$. By definition there exists a covering $\{U_i \to U\}$ such that $E|_{U_i}$ is represented by a strictly perfect complex. Thus E is pseudo-coherent (i.e., m-pseudo-coherent for all m) by Lemma 08CC. Moreover, a direct summand of a finite free module is flat, hence $E|_{U_i}$ has finite Tor dimension by Lemma 0654. Thus (2) holds.

Assume (2). Let U be an object of $\mathcal{C}$. After replacing U by the members of a covering we may assume there exist integers $a \leq b$ such that $E|_U$ has tor amplitude in $[a, b]$. Since $E|_U$ is m-pseudo-coherent for all m we conclude using Lemma 08CP. □